



Wearable devices could now be powered by your sweat



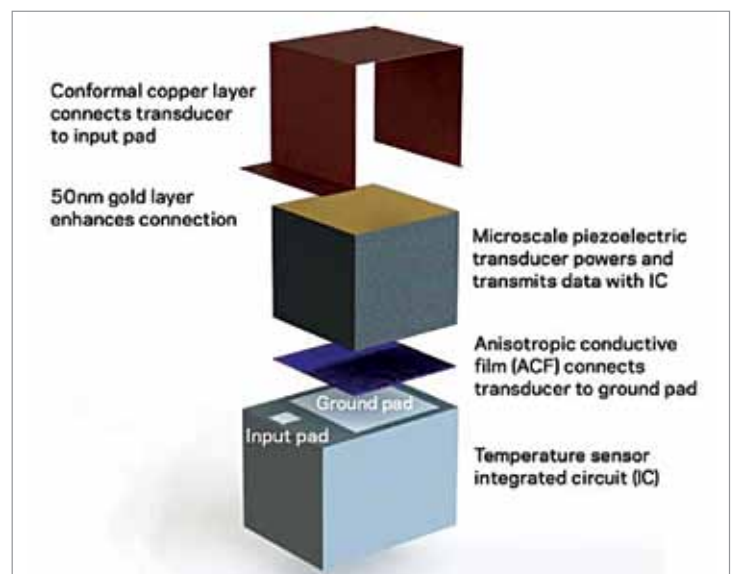
Wearable biofuel cells that produce electricity from sweat (Credit: www.tus.ac.jp/en)

A team of scientists from the Tokyo University of Science, Japan, are exploring efficient ways of using sweat as the sole source of power for wearable electronics. Recently, a novel design for a biofuel cell array has been presented that uses lactate (a chemical found in sweat) to generate enough power to drive a biosensor and wireless communication devices for a short time. The paper-bandage type device can be entirely fabricated via screen printing and can be worn. For example, whereas similar previous cells used silver wires as conducting paths, the present biofuel cells employ porous carbon ink. Paper layers are used for collecting sweat and simultaneously transport it to all cells through capillary effect—the same effect by which water quickly travels through a napkin when it comes into contact with a water puddle.



Tiny injectable chips to monitor body parameters

Implantable medical devices are transforming healthcare and improving the quality of life for millions of people by monitoring physiological conditions such as temperature, blood pressure, glucose, and respiration for both diagnostic and therapeutic procedures. Now researchers at Columbia University have built the smallest single-chip system, consuming a total volume of less than 0.1mm^3 . The system is as small as a dust mite and visible only under a microscope. In order to achieve this, the team used ultrasound to both power and communicate with the device wirelessly, since wavelengths for ultrasound are much smaller at a given frequency because the speed of sound is much less than the speed of light. The current device measures body temperature but there are many more possibilities.



Schematic representation of the device
(Credit: www.engineering.columbia.edu)